

FLL Challenge Judging Masterclass

Strategic Rubric Justification Homework — 10-Question Scenario Gauntlet

Hey RoboTigers and Lego Legends! When you are inside the FLL judging room, the judges use specific questions from their official handbook to grade your rubrics. For every scenario below, **both Option A and Option B are true things a team could say.**

- **Step 1:** Read the scenario and the official question carefully.
- **Step 2:** Circle the option (A or B) that you think strategically highlights our rubrics and Core Values best.
- **Step 3:** Write a 1-2 sentence defense explaining why your choice makes the strongest case to the judges.

PART 1: INNOVATION PROJECT

QUESTION 1 — RUBRIC SECTION: IDENTIFY

Our Team's Journey: Early in the season, the team spent three weeks researching solar panels. But during a group review, we realized solar panels didn't solve a unique, specific problem in our local neighborhood. The team decided to shift gears and research local neighborhood flash flooding instead, discovering that 15 homes on our school street flood every spring.

Official Judge Question: "Describe the problem you were trying to solve and how it connects to the project prompt."

Option A: "We started with solar energy but realized it wasn't a good project topic for us. We threw away those notes and immediately started a new step-by-step process: researching flood zones, surveying neighbors, and building our current prototype."

Option B: "We originally researched solar panels, but realized they didn't address an immediate local need. We practiced **Inclusion** and consensus to pivot toward local flash flooding, identifying a targeted problem where seasonal runoff explicitly damages 15 homes on our school's street."

I chose Option ____ because:

QUESTION 2 — RUBRIC SECTION: DESIGN

Our Team's Journey: When it came time to sketch out our project ideas, half the team wanted to build a mobile app and the other half wanted to build a physical community display board. Instead of arguing or letting one side outvote the other, the team listed the core benefits of both and designed an interactive kiosk board that connects directly to a mobile layout via QR codes.

Official Judge Question: "How did your team divide the work for your innovation project? ... How did you make sure everyone's work was valued?"

Option A: "We split our team into specialized pairs based on individual strengths: research, layout design, and expert outreach. When we hit a split design choice between an app and a board, we made sure everyone felt valued by integrating the best features of both into an interactive kiosk layout."

Option B: "We had an even split on what to build, so we held a vote. The app won, but to make sure the board group felt valued, we let them pick the color scheme and write the text descriptions for the app's home page."

I chose Option ____ because:

QUESTION 3 — RUBRIC SECTION: CREATE

Our Team's Journey: Our project solution is a smart recycling bin that uses an internal weight sensor to detect if a heavy glass bottle or dense object was placed in the paper-only compartment. Standard public recycling containers rely completely on passive sticker labels that people often ignore.

Official Judge Question: "What is innovative about your solution? ... What makes your idea different from other solutions you discovered in your research?"

Option A: "Our design stands out because it is built from highly durable, water-resistant recycled plastic panels and features a bright color scheme with custom illustrations to ensure it looks distinct in a crowded school hallway."

Option B: "Our **Innovation** is moving from passive education to active feedback. Existing solutions rely entirely on labels people overlook. Our system is different because it uses an integrated weight threshold sensor to provide immediate structural detection and alerts when a sorting mistake happens."

I chose Option ____ because:

QUESTION 4 — RUBRIC SECTION: ITERATE

Our Team's Journey: We brought our initial cardboard model of a community water filter to a city environmental engineer. They looked at it and noted that falling leaves and thick twigs would clog our fine internal mesh screen within 24 hours. The team took that feedback, returned to the workshop, and added an angled, coarse outer plastic debris shield.

Official Judge Question: "Who did you share your solution with for feedback? ... What changes or improvements did you make to your project throughout the process?"

Option A: "We presented our layout to a professional civil engineer. To **Iterate** based on their real-world warning, we redesigned the intake to feature a dual-stage system: an external debris shield to block large organic waste, which directly improved our operational lifespan."

Option B: "We met with a local engineer who liked our core concept but pointed out that leaves would break it. We modified our project by adding a strong plastic screen over the top so that the cardboard model underneath wouldn't get damaged or wet during testing."

I chose Option ____ because:

QUESTION 5 — RUBRIC SECTION: COMMUNICATE

Our Team's Journey: The team scheduled a presentation with the local library board to share our project. Right as we stood up, the library's projector went completely blank and our slide deck wouldn't load. Instead of panicking, the team immediately shifted to using our hand-drawn engineering notebook sketches and shared our project with clear, loud voices.

Official Judge Question: "What are you most proud of in regards to your project?"

Option A: "We are proudest of our **Resilience** and communication depth. When our digital slides crashed during a live library board meeting, we didn't give up. We communicated our ideas effectively using raw technical sketches, proving our team can handle high-pressure environments."

Option B: "We are most proud that our final presentation script is fully memorized by every single teammate, meaning we don't need to look at any cue cards or look down at the floor when we talk to judges or community leaders."

I chose Option ____ because:

PART 2: ROBOT DESIGN

QUESTION 6 — RUBRIC SECTION: IDENTIFY

Our Team's Journey: On day one of looking at the complex game mat, the team realized trying to tackle all 15 missions randomly would be overwhelming. We decided to map the points-per-second value of each task and noticed that 4 distinct missions clustered tightly together in the top corner, which could be reached using a single strategic launch path.

Official Judge Question: "How did your team decide which missions to work on?"

Option A: "We prioritized missions based on structural location and time efficiency. By executing a thorough **Discovery** phase, we targeted a cluster of 4 high-value missions in the top-left quadrant, allowing us to maximize field points while conserving battery life and clock time."

Option B: "We voted on which missions looked the coolest and most exciting to watch. We chose the top corner tasks because they had moving plastic pieces that our lifting arm could easily push over without getting stuck."

I chose Option ____ because:

QUESTION 2 — RUBRIC SECTION: DESIGN

Our Team's Journey: Our team has two primary programmers who spend most of their time working on the laptop software. To ensure the builders understood the underlying logic, the programmers organized weekly code walk-throughs, demonstrating how sensor thresholds dictate motor movements.

Official Judge Question: "How did your team divide the work on your robot? ... How did you make sure every team member understood the robot and coding?"

Option A: "The builders ran the hardware assembly while the programmers controlled the code. To maintain absolute **Inclusion**, we established 'peer calibration' sessions where programmers explained the logic blocks to the builders, ensuring everyone can explain the code during judging."

Option B: "We split the tasks completely down the middle. The programmers wrote the code and then printed it out in a binder so the builders could look at it and read the comments whenever they had free time during our meetings."

I chose Option ____ because:

QUESTION 3 — RUBRIC SECTION: CREATE

Our Team's Journey: To complete a precise 90-degree turn, our robot originally relied on raw time-duration blocks. However, minor shifts in battery levels caused the turns to drift. We updated the system to incorporate a Gyro Sensor, developing a reusable custom block ("My Block") that continuously monitors live angular data to stop the motors.

Official Judge Question: "Explain how you used sensors and coding to complete missions. ... What is innovative about your robot attachments or coding?"

Option A: "Our code is highly innovative because it is extremely long and detailed, taking up over three full screens of scrolling blocks, which proves we spent a massive amount of hours perfecting our navigation script."

Option B: "We achieved coding precision by creating an autonomous closed-loop turn system. Our **Innovation** is a custom Gyro-steering 'My Block' that reads real-time sensor variables. Instead of guessing seconds, the robot constantly reads sensor updates to halt at exactly 90 degrees."

I chose Option ____ because:

QUESTION 4 — RUBRIC SECTION: ITERATE

Our Team's Journey: The team built an expansive motorized lift arm to capture a heavy game piece. During initial testing, the arm worked perfectly on some runs but dropped the piece completely on others. We started running structured 10-match trial sets, discovered a 60% failure rate due to slick plastic-on-plastic contact, and resolved it by adding rubber bands to the claw grip.

Official Judge Question: "How did you test that your robot was consistent at solving a mission? ... Describe how you improved your robot after testing."

Option A: "We tested the arm over and over again on the table until it finally worked a couple of times in a row. Once we saw it complete the mission successfully, we knew our robot was ready for tournament day."

Option B: "We tracked consistency quantitatively by executing sets of 10 controlled field trials. After tracking a 60% mechanical failure rate, we used the engineering design cycle to **iterate**, integrating high-friction rubber band contact points that raised consistency to 90%."

I chose Option ____ because:

QUESTION 5 — RUBRIC SECTION: COMMUNICATE

Our Team's Journey: At the conclusion of building a compact, modular robot chassis, the team realized that our true accomplishment wasn't just individual mechanisms, but how seamlessly our different design ideas attached together onto a single common motor drive base.

Official Judge Question: "What are you most proud of with regards to your robot design or mission strategy?"

Option A: "We are most proud that our robot is faster than last year's model, allowing us to hit a peak score of 290 points during our best automated test run of the season."

Option B: "We are proudest of our modular engineering strategy. We developed a unified base chassis that allows quick-swap attachments to pin down securely in seconds. This highlights our **Teamwork**, as separate building ideas successfully integrated into one system."

I chose Option ____ because:
